ReRAPTOR Search Framework

Shared timetable-feasible search backbone with task-specific scoring and three-layer pruning

Step 1

Load Query

GTFS stop schedules,
origin, departure time,
time budget, and score profile.

Step 2

Initialize Label

Create the base label:
arrival time, visited-stop state,
counters, and leg history.

Step 3

Expand Timetable Edges

Enumerate feasible outgoing
services under transfer
and budget constraints.

Step 4

Update Route State

Refresh ride time,
unique stops, geometry features,
and task-specific accumulators.

Step 5

Compute Search Score

Rank each partial route
with a task-aware surrogate score
before acceptance.

Three-Layer Pruning

NEG Reject loops and clearly weak partial routes.

DOM Discard labels dominated at the same stop and round.

BEAM Keep only the top-W labels per stop-round bucket.

Step 6

Keep Accepted Labels

Store feasible survivors
for final ranking
and route reconstruction.

Step 7

Return Final Itinerary

Iterate until the frontier is exhausted
or a resource cap is reached,
then output the highest-scored accepted route.

Shared search backbone

Steps 1 to 4 are unchanged
across the experiments.

Task-specific modules

Scoring, pruning predicates,
and acceptance rules change
by experiment.